

Pattern Matching and Machine Learning for Audio Signal Processing

Exercise sheet 6

To be uploaded in eCampus till: 21-06-2026 22:00 (strict deadline)

Exercise 6.1

[4 + 2 = 6 points]

Consider the convolution operator as defined in Chapter 3:

$$C_h[x](n) := (h * x)(n) = \sum_{k \in \mathbb{Z}} h(k)x(n - k).$$

- (a) How can C_h be realized as a matrix, such that $C_h[x]$ can be written as a matrix $\times$ vector product? Motivate your answer.
- (b) How can you modify the matrix in order to obtain a (finite dimensional) matrix for computing the cyclic convolution of finite signals of length N ?

Exercise 6.2

[4 points]

Calculate the frequency response of the following signal:

$$h(n) = \begin{cases} 1/3 & \text{if } n = 0, 1, 2 \\ 0 & \text{else} \end{cases}$$

Exercise 6.3

[4 + 4 = 8 points]

Investigate whether the following operators are time-invariant:

- (a) upsampling
- (b) frequency-shift operator.

Prove your claim.